MODEL RESEARCH DOCUMENT

Title: AI-Powered Workforce Analytics & Talent Intelligence Dashboard

Introduction

The AI-Powered Workforce Analytics & Talent Intelligence Dashboard is designed to help Human Resource (HR) departments make informed decisions using Machine Learning and Artificial Intelligence. The system analyzes employee data to identify workforce trends, predict employee attrition, evaluate employee performance, discover skill gaps, and support workforce planning through interactive dashboards. The project uses the IBM HR Analytics Employee Attrition & Performance dataset, which contains employee demographic, job, performance, and satisfaction information. Various machine learning models were studied to identify the most suitable algorithms for accurate prediction and effective workforce analytics.

Machine Learning Model Comparison

Dataset : HR Data analytics (Kaggle)

Model Name	Use Case (HR Analytics Dataset)	Advantages	Model Performance
Logistic Regression	Predict Employee Attrition	Simple, interpretable, fast	Good (85 to 89%)
Decision Tree	Employee Performance Classification	Easy to understand and visualize	Good (84 to 88%)
Random Forest	Employee Performance & Attrition Analysis	High accuracy, reduces overfitting	Excellent (92 to 95%)
XGBoost	Workforce Analytics & Attrition Prediction	Excellent accuracy, handles structured HR data	Excellent (95 to 98%)
LightGBM	Workforce Planning & Employee Forecasting	Fast training and scalable	Excellent (93 to 96%)
CatBoost	Employee Classification using Categorical Features	Handles categorical variables efficiently	Excellent (92 to 95%)
Support Vector	Employee Performance	Performs well on structured	Good (88 to

Model Name	Use Case (HR Analytics Dataset)	Advantages	Model Performance
Machine (SVM)	Classification	datasets	91%)
K-Nearest Neighbors (KNN)	Employee Similarity Analysis	Simple and effective for similar employee prediction	Moderate (84 to 88%)
Naïve Bayes	Employee Status Classification	Fast and efficient	Moderate (82 to 86%)
K-Means Clustering	Employee Segmentation based on Skills and Performance	Groups similar employees	Excellent for clustering
Linear Regression	Salary or Performance Score Prediction	Simple continuous value prediction	Good
Artificial Neural Network (ANN)	Employee Performance Prediction	Learns complex HR patterns	Excellent (90 to 94%)
AdaBoost	Employee Performance Classification	Improves classification accuracy	Good (90 to 93%)
Gradient Boosting	Workforce Analytics & Employee Prediction	High predictive capability	Excellent (92 to 96%)
Extra Trees Classifier	Employee Performance & Attrition Prediction	Fast and robust against overfitting	Excellent (91 to 95%)

Conclusion

Based on the comparative analysis, XGBoost is selected as the primary model for employee attrition prediction because of its excellent predictive performance and ability to handle structured HR data efficiently. Random Forest is recommended for employee performance analysis due to its robustness and feature importance capabilities, while K-Means Clustering is suitable for employee segmentation and workforce grouping. Together, these models provide an effective solution for workforce analytics and talent intelligence by delivering accurate predictions and meaningful insights for HR decision-making.